

Luís André Oliveira

Tocha - Coimbra, Portugal

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Education

University of Aveiro

MSc in Informatics Engineering

Aveiro, PT

Sept 2019 - 2021

University of Aveiro

BTech in Informatics Engineering

Aveiro, PT

Sept 2016 - 2020

Work Experience

Critical Software

Coimbra, PT

Software Engineer - Android

May 2024 - Current

- Development of the audio system of the car, with focus on the audio control and volume controls of the infotainment system of the car.
- Work in HAL, developed in C++, the first layer of communication with the hardware and where all the business logic is related to hardware interactions are implemented. Code tested with unit tests using GTest.
- Development of Android Services to provide APIs to the Android apps to interact with the audio settings, audio controls, and volume controls. Services developed using Kotlin and tested using mockito
- **Technical Skills:** Android, Kotlin, C++, AIDL, GMock, Mockito.

Software Engineer - NodeJS

November 2021 - May 2024

- Development of infotainment applications to cars.
- Typescript-based applications, developed using NodeJS, with interaction with several signals received from the car and communication with backends via REST APIs.
- Development of unit tests using JestJS and behavior verification using integration tests written with Gherkin.
- Start a new project from the beginning where I was an active part of finding requirements and designing the architecture of the app.
- Work in a Scrum team.
- **Technical Skills:** Typescript, NodeJS, RxJS, behavior-driven development using Cucumber. CI/CD using Jenkins and Github actions.

Projects

Air quality monitoring platform

Aveiro, PT

University of Aveiro - Master thesis

November 2020 - November 2021

- Aggregation of data collected by several air pollutant sensors with different data models in one common data model.
- Development of a web and mobile application to show the data collected with several ways to analyze the data (dashboards, raw data). The web application was developed using Django and the mobile application using Flutter
- Provide a Rest API so their clients can integrate the data on their solutions.
- Provide dashboards via Grafana to monitor the concentrations of the air pollutants.
- All system components were working on docker containers.
- Project developed in collaboration with the Environment Institute of the University of Aveiro.
- **Technical Skills:** Python with Django, HTML, CSS, JavaScript, PostgreSQL, Rest Api, Docker, Flutter, Grafana, RabbitMQ, OpenAPI.

Visualization tool for life logging

Aveiro, PT

University of Aveiro - Bachelor final project

February 2019 - June 2019

- System to improve the lifestyle based on statistics about daily activities identified by photo analysis of photos taken periodically by the smart-phone. Development of a categorization algorithm to extract information from the photos collected.
- Development of a web application to present the statistics of the activities and a mobile application to collect the photos.
- **Technical Skills:** Python with Django, HTML, CSS, JavaScript, and Java to develop the mobile app.

Skills and Interests

Programming Typescript/Javascript, NodeJS, ReactJS, NestJS, JestJS, Python, HTML/CSS, SQL.

Miscellaneous Linux, Git, Jenkins, Github Actions, Docker, Gherkin

Languages

English Professional proficiency

Portuguese Native proficiency